

Vol 13 Chapter 5 — Evolution as Substrate Dynamics

Keeper (author pass — deep math/physics revision)

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Chapter 5 — Evolution as Substrate Dynamics

Level 1 — one sentence

Darwinian evolution — variation (mutation) + heredity (DNA replication) + selection (differential survival/reproduction) → adaptation — is the population-level dynamics of substrate K-type configurations across biological time, with mutations as substrate K-type alterations and selection as substrate-permeated environmental filtering.

Level 2 — graduate-physicist precision

5.1 Darwin's mechanism

Darwin 1859, *Origin of Species*: natural selection operates on heritable variation. Modern synthesis (1930s-40s, Fisher-Haldane-Wright): mathematical population genetics integrates Mendel + Darwin.

Selection equation (continuous, single locus):

$$\frac{dp}{dt} = sp(1 - p)$$

with p allele frequency, s selection coefficient. Fixation time $\sim 1/s$ generations for positive s .

5.2 Mutation rates

Per-base-pair per generation: - Bacteria (*E. coli*): $\sim 10^{-9}$ - Mammals: $\sim 10^{-8}$ - Viruses (RNA): $\sim 10^{-4}$

Genome-wide: ~ 1 mutation per replication for bacteria, ~ 100 for mammals.

5.3 Genetic drift

Random sampling in finite populations: drift dominates when $|s| < 1/(2N_e)$ with N_e effective population size.

Neutral theory (Kimura): most molecular evolution is drift, not selection. Molecular clock = drift accumulating mutations.

5.4 Substrate dynamics view

Substrate provides: - Mutation: substrate K-type alteration (radiation, chemical damage, replication error) - Selection: substrate-mediated environment filters K-type configurations - Drift: substrate stochasticity at sampling

BST observation: substrate dynamics include both deterministic (selection on K-type stability) and stochastic (drift, mutation) components — same epistemic-stochasticity reading as DCCP (memory).

5.5 Phylogenetics

Reconstruct evolutionary history from molecular sequences. Tree-of-life topology: 3 domains (Bacteria, Archaea, Eukarya) with horizontal gene transfer making the tree more of a web.

Common ancestor: LUCA ~3.5 Gya. Eukaryotes ~1.8 Gya. Animals ~700 Mya. Vertebrates ~500 Mya. Primates ~85 Mya. *Homo sapiens* ~300 Kya.

5.6 K-audit anchors

- Vol 6 Ch 1-4 (statistical mechanics ↔ population genetics)
- Vol 14 (information theory ↔ evolution)
- DCCP (memory; epistemic-deterministic substrate dynamics)

Level 3 — 5th-grader accessibility

Evolution = variation + heredity + selection → adaptation. Darwin 1859. Modern math: $dp/dt = sp(1 - p)$. Mutation rate: 10^{-9} to 10^{-4} per base per generation. Drift vs selection: stochastic vs deterministic. In BST: mutations are substrate K-type alterations; selection is substrate-environmental filtering; same epistemic-stochasticity as DCCP. LUCA ~3.5 Gya. Humans ~300 Kya.

What comes next

Chapter 6 develops biochemistry bridge to Vol 12 Chemistry.

Where to look this up

- Darwin 1859
- Fisher-Haldane-Wright (Modern Synthesis)
- BST: Vol 6, Vol 14, DCCP